E_t = true value − approximation

true error

$$\varepsilon_t = \frac{\text{true error}}{\text{true value}} \cdot 100\,\%$$

↳ true percent relative error

$$\varepsilon_a = \frac{\text{approximate error}}{\text{approximation}}\, 100\,\%$$

approximate value

$|\varepsilon_a| < \varepsilon_s \quad \varepsilon_s = (0.5 \times 10^{2-n})\,\%$

$(-1)^s \times (1.f)_2 \times 2^{\text{exponent}-127}$

e → exponent

mb → base of the number system used

↳ mantissa

$\frac{1}{b} \le m < 1 \quad \varepsilon_m = b^{1-t}$

↳ machine epsilon

$1 + \delta = 1$, whenever $|\delta| < \varepsilon_m$

Taylor's Thm:

$$f(x) = f(a) + f'(a)(x-a) + \frac{1}{2!}f''(a)(x-a)^2 + \frac{1}{3!}f'''(a)(x-a)^3 + \frac{1}{n!}f^{(n)}(a)(x-a)^n + R_n$$

(x_{i+1} ↓ , x_i ↓)

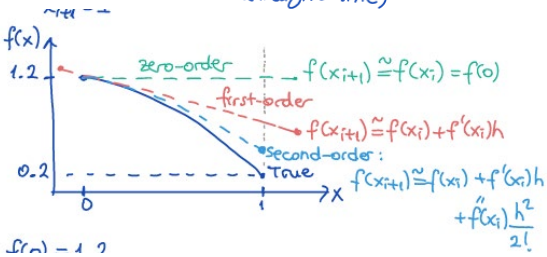
$$R_n = \int_a^x \frac{(x-t)^n}{n!} f^{(n+1)}(t)\,dt$$, t is a dummy variable.

(If and its first (n+1) derivatives are continuous on an interval containing a and x.)

Term by term

First term $f(x_{i+1}) \cong f(x_i)$ → zero-order approximation (constant)

$f(x_{i+1}) \cong f(x_i) + f'(x_i)(x_{i+1} - x_i)$ → first order approximation (straight line)



Solution: For $h = 0.5$

$x_{i-1} = 0$, $f(x_{i-1}) = 1.25 = f(0)$

$x_i = 0.5$, $f(x_i) = 0.825 = f(0.5)$

$x_{i+1} = 1$, $f(x_{i+1}) = 0.25 = f(1)$

Forward diff:

$$f'(0.5) \cong \frac{f(1) - f(0.5)}{1 - 0.5} = \frac{0.2 - 0.925}{0.5} = -1.45$$

$|\varepsilon_t| = 58.9\,\%$

Ex: Given $\tilde{x} = 2.5$ with an error of $\Delta\tilde{x} = 0.01$. Estimate the resulting error in the function $f(x) = x^3$

$$\Delta f(\tilde{x}) = |f'(\tilde{x})|\,\Delta\tilde{x} \cong \left|3\tilde{x}^2\,\Delta\tilde{x}\right|_{\tilde{x}=2.5} = 3(2.5)^2(0.01) = 0.1875$$

The relative error of x is given by $\frac{x - \tilde{x}}{x}$

Condition number = $\frac{\text{Relative error of } f(x)}{\text{Relative error of } x} = \frac{\tilde{x} f'(\tilde{x})}{f(\tilde{x})}$

Bisection Method

Choose x_l and x_u s.t. $f(x_l)f(x_u) < 0$

Estimate root as:

$$x_r = \frac{x_l + x_u}{2}$$

$$\varepsilon_a = \left|\frac{x_r^{new} - x_r^{old}}{x_r^{new}}\right| \times 100\,\%$$

if $\varepsilon_a < \varepsilon_{pr}$ stop)

The False-Position Method

$$x_r = x_u - \frac{f(x_u)(x_l - x_u)}{f(x_l) - f(x_u)}$$

Simple Fixed Point Iteration

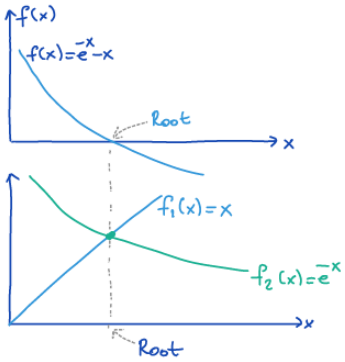
$x = g(x)$ (1) $x_{i+1} = g(x_i)$

$$\varepsilon_a = \left|\frac{x_{i+1} - x_i}{x_{i+1}}\right| 100\,\%$$

$e^{-x} - x = 0$ Determine the roots graphically.

Reformulate as:

$y_1 = x$, $y_2 = e^{-x}$; $x = e^{-x}$ (y_1 = $f_1(x)$, $y_2 = f_2(x)$)



Converges if $|g'(x)| < 1$ around the root

(Derivation in the book)

The Newton-Raphson Method

$$x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}$$

$$\varepsilon_{t,i+1} = -\frac{f''(x_r)}{2f'(x_r)}\varepsilon_{t,i}^2$$

true error at iteration (i+1); ↳ true root

The Secant Method

$$x_{i+1} = x_i - \frac{f(x_i)(x_{i-1} - x_i)}{f(x_{i-1}) - f(x_i)}$$

Modified Secant Method

δ : a small perturbation fraction

$$x_{i+1} = x_i - \frac{\delta x_i\, f(x_i)}{f(x_i + \delta x_i) - f(x_i)}$$

Brent's Method

Three ptr: (x_{i-2}, y_{i-2}), (x_{i-1}, y_{i-1}) and (x_i, y_i)

A quadratic function of y

$$g(y) = \frac{(y-y_{i-1})(y-y_i)}{(y_{i-2}-y_{i-1})(y_{i-2}-y_i)} x_{i-2} + \frac{(y-y_{i-2})(y-y_i)}{(y_{i-1}-y_{i-2})(y_{i-1}-y_i)} x_{i-1} + \frac{(y-y_{i-2})(y-y_{i-1})}{(y_i-y_{i-2})(y_i-y_{i-1})} x_i$$

The root x_{i+1} corresponds to $y = 0$

$$x_{i+1} = \frac{y_{i-1}\,y_i}{(y_{i-2}-y_{i-1})(y_{i-2}-y_i)} x_{i-2} + \frac{y_{i-2}\,y_i}{(y_{i-1}-y_{i-2})(y_{i-1}-y_i)} x_{i-1} + \frac{y_{i-2}\,y_{i-1}}{(y_i-y_{i-2})(y_i-y_{i-1})} x_i$$

$y = f(x)$ and $x = g(y)$ forms.